UNIVERSITY OF AGRICULTURAL SCIENCES, BENGALURU

COLLEGE OF AGRICULTURE , GKVK – 560065

PROJECT REPORT

EAI - 421 : Application Of AI & ML in Agriculture (0+10)

Project Title : Application of Artificial Intelligence and Machine Learning for Groundwater Quality

TEAM : AMB2016-AMRUTHA VARSHINI

AMB2103-K.KAVYA

AMB2177-PROMOD.P

SUBMITTED TO:

DR.PRABHURAJ METHIPATIL

COURSE TEACHER AI &ML IN AGRICULTURE

COA,GKVK-BENGALURU

Content

INTRODUCTION

Groundwater is a critical resource for drinking, irrigation, and industrial applications, but its quality is increasingly impacted by natural processes and human activities.
Timely and accurate assessment of groundwater quality is essential for sustainable management.
This capstone project presents the development of an Artificial Intelligence (AI) and Machine Learning (ML)-based

system to analyze and predict groundwater quality using realworld datasets, enhancing the efficiency of traditional evaluation methods.

PROBLEM STATEMENT

Groundwater quality assessment is crucial for safe drinking water and agriculture.

Traditional methods are time-consuming, costly, and lack real-time insights.

2018 data shows variations in pH, EC, TDS, hardness, indicating health and productivity risks.

Need for efficient, accurate, and scalable AI-driven system to predict water quality.

Project Goal

Develop ML model to classify groundwater as safe/unsafe, improving prediction accuracy and decision-making.

MOTIVATION

Importance

- Ensures safe drinking water (health risks: fluoride, arsenic, nitrate)
- Protects agriculture productivity & soil health
- Addresses India's water scarcity & stress

Practical Applications

- Precision irrigation & crop selection
- Early contamination alerts for farmers
- Informed policy & remediation efforts

Relevance

- Supports Clean Water
- Water management, agri-tech, risk assessment

OBJECTIVES

1.To analyze the 2018 post-monsoon groundwater quality dataset and understand variations in key physicochemical parameters such as pH, EC, TDS, and hardness.

2. To develop and apply machine learning models (Random Forest, Decision Tree, etc.) for predicting and classifying groundwater quality.

3. To evaluate and compare model performance to identify the most accurate approach for groundwater quality prediction.

DATASET OVERVIEW

Dataset source: Ground water quality 2018 post monsoon data from Kaggle

Number of records and features: 300 samples, 9 features (pH, EC, TDS, Hardness, Alkalinity, Chloride, Sulphate, Nitrate, Fluoride)

Types of features: Numerical (physicochemical parameters)

Preprocessing steps: Data loading, feature selection, normalization, train-test split

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FEATURE	Type	Source
Ph	Numerical (Continuous)	Field measurement
Electrical Conductivity	Numerical (Continuous)	Field measurement
Total dissolved solids	Numerical (Continuous)	Field measurement
FEATURE	Type	Source
Total hardness	Numerical (Continuous)	Field measurement
Alkalinity	Numerical (Continuous)	Field measurement
Chloride	Numerical (Continuous)	Laboratory analysis

Sulphate	Numerical (Continuous)	Laboratory analysis
Nitrate	Numerical (Continuous)	Laboratory analysis
Fluoride	Numerical (Continuous)	Laboratory analysis
Groundwater Quality Index	Numerical (Target)	Computed/Derived

APPROACH

Data Preprocessing

- Upload CSV/Excel dataset in Google Colab
- Feature selection (pH, EC, TDS, etc.)
- Normalize data using MinMaxScaler
- Train-test split (80:20)

Exploratory Data Analysis

- Water quality distribution
- Correlation heatmap
- Identify key influencing parameters

Model Selection

- Random Forest
- Decision Tree
- Logistic Regression

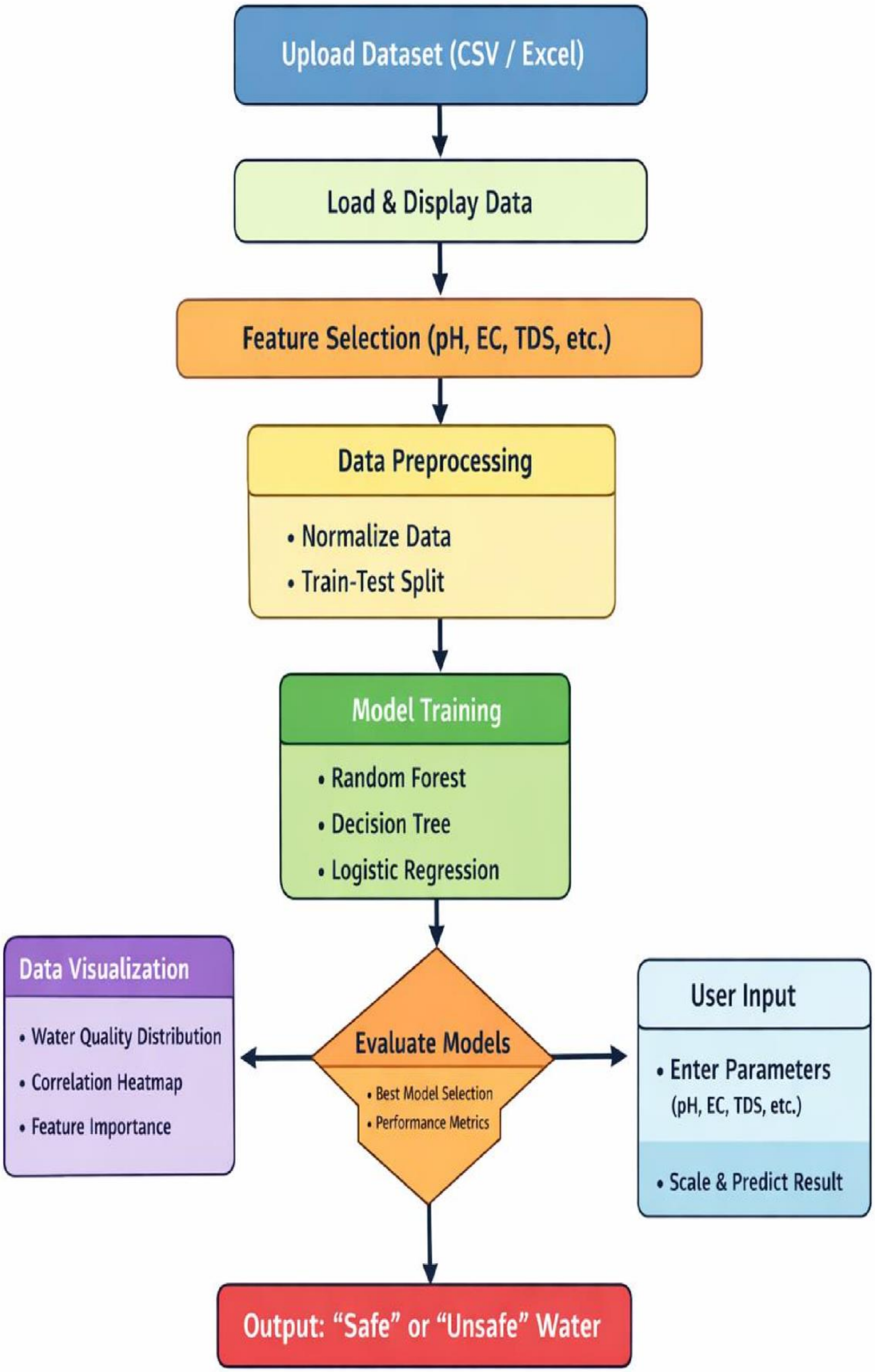
Model Training & Validation

- Train models on training data
- Test on unseen data
- Select best-performing model

Evaluation Metrics

- Accuracy
- F1-score
- Confusion Matrix

FLOW CHART


Upload Dataset (CSV / Excel)
Load & Display Data
Feature Selection (pH, EC, TDS, etc.)
Data Preprocessing
• Normalize Data
• Train-Test Split
Model Training
• Random Forest
• Decision Tree
• Logistic Regression
Data Visualization
• Water Quality Distribution
• Correlation Heatmap
• Feature Importance
Evaluate Models
• Best Model Selection
• Performance Metrics
User Input
• Enter Parameters
(pH, EC, TDS, etc.)
• Scale & Predict Result
Output: "Safe" or "Unsafe" Water

TOOL/TECHNOLOGY	Role in the Project
Python 3.x	Core programming language used to implement the entire ML pipeline
Pandas	Used for data loading, manipulation, and preprocessing of dataset
NumPy	Performs numerical computations and array operations
matplotlib	Used for data visualization like bar charts and heatmaps
scikit-learn	Machine learning library used for model building and evaluation
Random Forest Classifier	Ensemble algorithm that improves accuracy using multiple decision trees
Decision Tree Classifier	Tree-based model that classifies data using decision rules
Logistic Regression	Statistical model used for binary classification (Safe/Unsafe)
train_test_split	Splits dataset into training and testing sets
Confusion Matrix	Evaluates classification performance by comparing actual vs predicted values

PYTHON CODE

```
--- 1. Upload File in Colab ---  from
google.colab import files
uploaded = files.upload()

# Get uploaded file name file_name
= list(uploaded.keys())[0]

# --- 2. Import Libraries ---
import numpy  as np import
pandas as pd import
matplotlib.pyplot as plt

from sklearn.ensemble import RandomForestClassifier from
sklearn.tree import DecisionTreeClassifier from
sklearn.linear_model import LogisticRegression from
sklearn.preprocessing import MinMaxScaler from
sklearn.model_selection import train_test_split from
sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

import warnings
warnings.filterwarnings('ignore')

# --- 3. Load Dataset ---
```

```
if file_name.endswith('.csv'):
df = pd.read_csv(file_name)
else:
   df = pd.read_excel(file_name)

print("Dataset Loaded:\n", df.head())

# --- 4. Feature Selection ---
features = ['pH', 'EC', 'TDS', 'Alkalinity', 'Hardness', 'Cl', 'F', 'NO3',
'SO4']

X = df[features] y =
df['Water_Quality']

# --- 5. Preprocessing --- scaler
= MinMaxScaler()
X_scaled = scaler.fit_transform(X)

# --- 6. Train-Test Split ---
X_train, X_test, y_train, y_test = train_test_split(
   X_scaled, y, test_size=0.2, random_state=42
)

# --- 7. Model Training ---
models = {
   "Random Forest": RandomForestClassifier(),
```

```
    "Decision Tree": DecisionTreeClassifier(),
    "Logistic Regression": LogisticRegression()
}

results = {}

for name, model in models.items():
model.fit(X_train, y_train)     y_pred
= model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
results[name] = acc

    print(f"\n{name} Results:")
print("Accuracy:", acc)
    print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification     Report:\n",     classification_report(y_test,
y_pred))

# --- 8. Best Model ---
best_model = max(results, key=results.get) print("\nBest
Model:", best_model)

# --- 9. Visualization ---

# 1. Water Quality Distribution plt.figure()
```

```
df['Water_Quality'].value_counts().plot(kind='bar')
plt.title("Water Quality Distribution") plt.show()

# 2. Correlation Heatmap
plt.figure() corr =
df.corr()
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Heatmap") plt.show()

# 3. Feature Importance best =
models[best_model] importances =
best.feature_importances_

plt.figure()
plt.bar(features, importances) plt.xticks(rotation=45)
plt.title("Feature Importance (" + best_model + ")") plt.show()

# --- 10. User Input for Prediction --- print("\nEnter
Water Quality Parameters:")

ph = float(input("pH: ")) ec =
float(input("EC: ")) tds =
```

float(input("TDS: ")) alk =

float(input("Alkalinity: ")) hard =

float(input("Hardness: ")) cl =

float(input("Cl: ")) f =

float(input("F: ")) no3 =

float(input("NO3: ")) so4 =

float(input("SO4: "))

sample = [[ph, ec, tds, alk, hard, cl, f, no3, so4]] sample_scaled

= scaler.transform(sample)

prediction = models[best_model].predict(sample_scaled)

print("\nPrediction:", "Safe Water" if prediction[0] == 1 else "**Unsafe Water")**

"Safe Water" if prediction[0] == 1 else "**Unsafe Water")**

Random Forest Results:

Accuracy: 0.95. Confusion Matrix:[[57 0] [3 0]] Classification Report:
precision recall f1-score support
0 0.95 1.00 0.97
57 1 0.00 0.00 0.00
3.accuracy - 0.95
60macro avg 0.47 0.50 0.49 60weighted avg 0.90
0.95 0.93 60

Decision Tree Results:

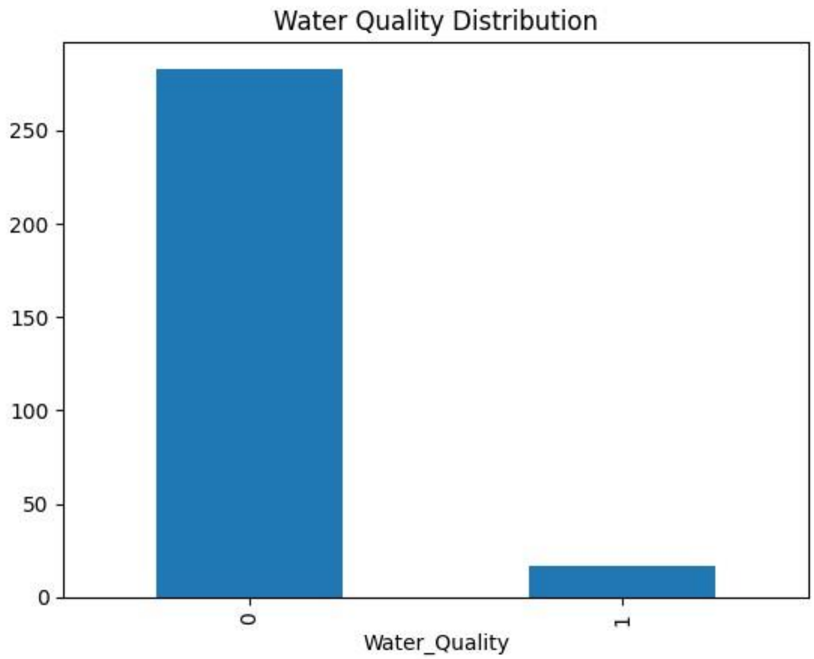
Accuracy: 0.9666666666666667
Confusion Matrix:[[57 0] [2 1]]
Classification Report:
precision   recall f1scoresupport
0   0.97   1.00   0.98   57   1   1.00   0.33   0.50
3.accuracy -0.97
60macro avg   0.98   0.67   0.74
60weighted avg   0.97   0.97   0.96   **Logistic Regression**

Results:

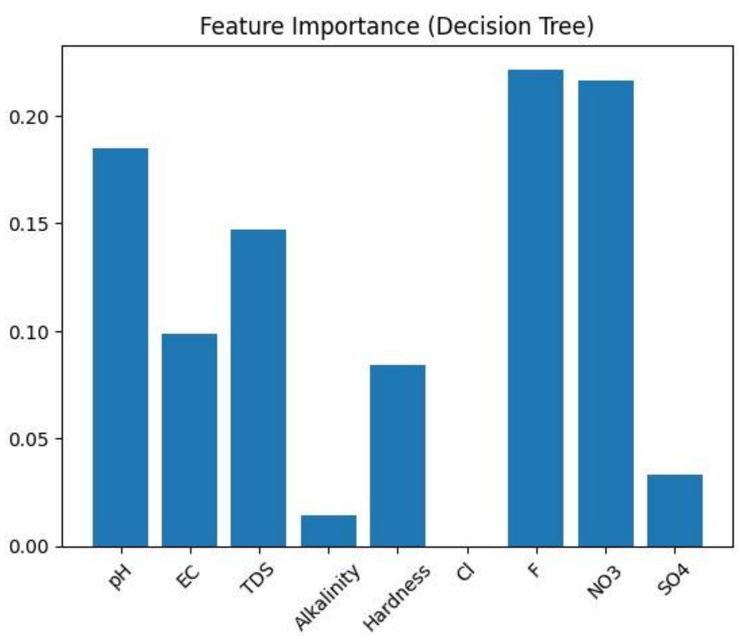
Accuracy: 0.95
Confusion Matrix:[[57 0] [3 0]]
Classification Report:
precision   recall f1-score   support   0   0.95   1.00   0.97
57   1   0.00   0.00   0.00
3.accuracy - 0.95
60macro avg   0.47   0.50   0.49   60weighted avg   0.90
0.95   0.93   60

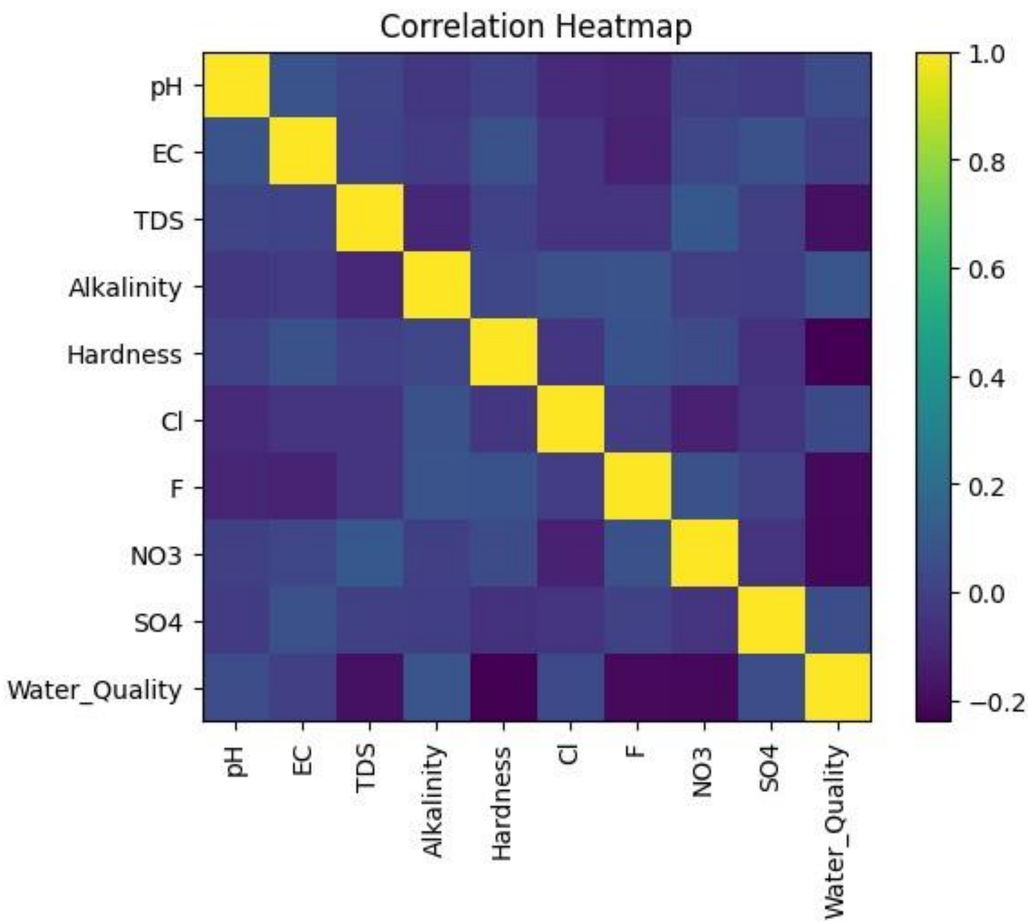
Best Model: Decision Tree

OUTPUT


Water Quality Distribution
250
200
150
100
50
0
0
1
Water_Quality

•


Feature Importance (Decision Tree)
0.20
0.15
0.10
0.05
0.00
pH
EC
TDS
Alkalinity
Hardness
Cl
F
NO3
SO4



DISCUSSION

Key Insights Derived from the Project

1. High accuracy in classifying groundwater quality based on physicochemical parameters.
2. Parameters like TDS, EC, and hardness had strong influence on water quality prediction.

Challenges Faced (Data Quality, Noise, Imbalance, Feature Selection)

1. Presence of missing values and noisy data affected model performance.
2. Class imbalance (more “safe” than “unsafe” samples) required careful handling for accurate prediction.

Limitations of the Approach or Dataset

1. Limited dataset size and region-specific data reduce generalization to other areas.
2. Lack of temporal data (seasonal variations) limits longterm prediction capability.

Impact on Agricultural Decision-Making

1. Helps farmers identify safe groundwater for irrigation, improving crop yield and soil health.
2. Supports efficient water resource management and sustainable farming.

CONCLUSION & FUTUREWORK

Project Summary & Outcomes Random Forest performed best, providing reliable and accurate predictions.

Potential Improvements Use larger and more diverse datasets (multi-seasonal, multi-location).
Apply advanced models (Deep Learning, Ensemble techniques) and real-time IoT-based monitoring systems.

Practical Applications in Farming Helps farmers select suitable water for irrigation, preventing soil degradation. Supports precision agriculture and sustainable water management.

Learning & Acknowledgement Gained hands-on experience in data preprocessing, model building, and evaluation. Developed skills in applying AI/ML for real-world agricultural problems.

REFERENCES

DATA SOURCE: Ground water quality 2018 post monsoon data from Kaggle

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   - *Research Journal of Engineering Sciences* ISSN 22789472, 2014
2. Praharsh S Patel, Dishant M Pandya, Manan Shah (2023)
   - *Environmental Science and Pollution Research*, 30(19), 54303-54323, 2023
3. Joshua O Ighalo, Adewale George Adeniyi, Gonçal
   - *Modeling Earth Systems and Environment*, 7(2), 669–681, 2021
